

1 Nash Equilibrium Strategy Profile

Having established the equilibrium structure in Section ?? and derived the indifference equations in Section ??, we now present the complete solution. Figure 1 displays the Nash equilibrium strategy profiles for various betting limits, ranging from lenient ($L = 0, U = 10$) to restrictive ($L = 0.5, U = 1$).

The system of equations from Section ?? was solved symbolically using SymPy (see Appendix ??), yielding closed-form expressions for all threshold values and strategic functions.

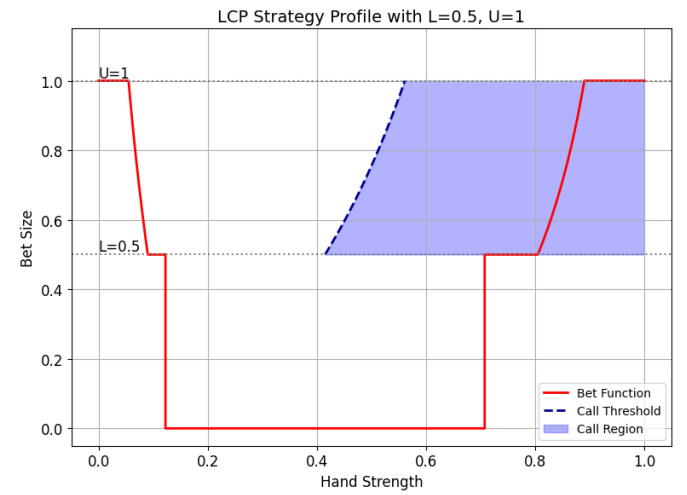
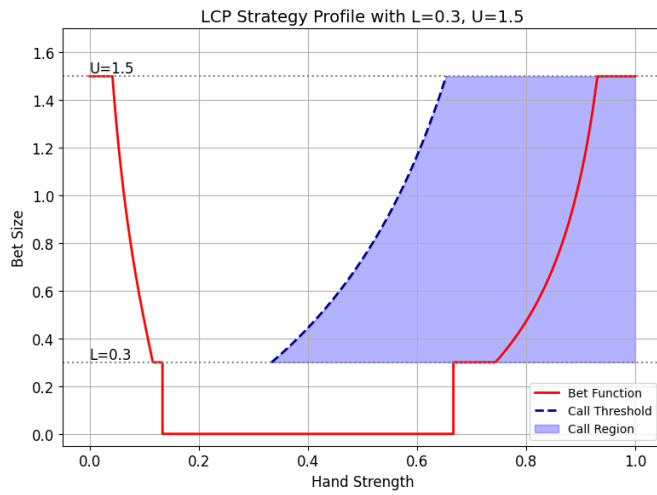
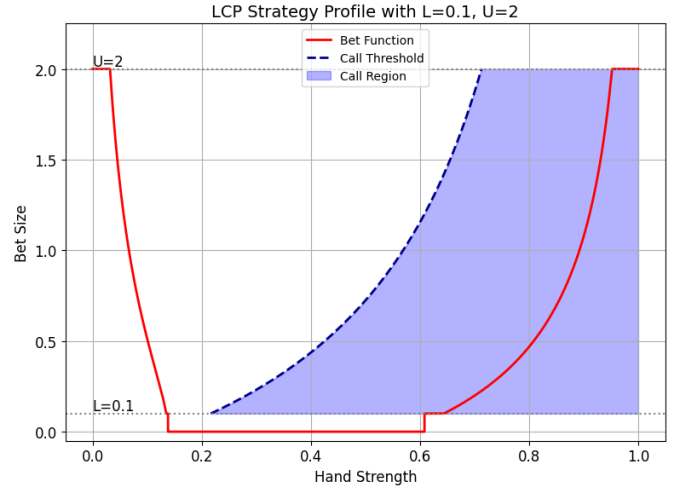
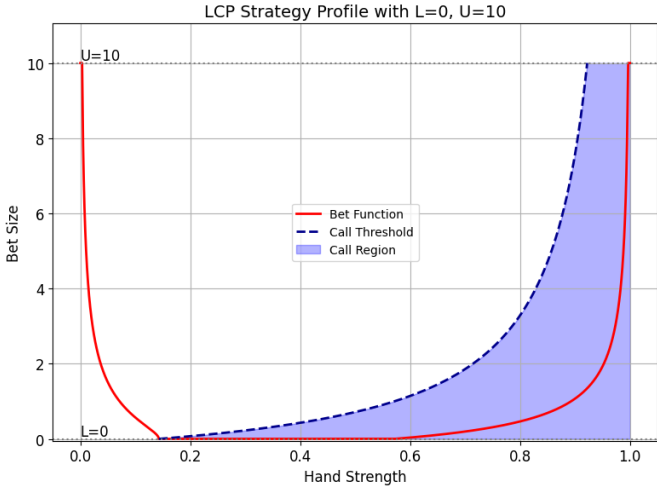


Figure 1: Nash equilibrium strategy profiles for different values of L and U , from lenient to restrictive bet size limits. The bet function maps hand strengths to bet sizes, while the call function gives the minimum calling hand strength for a given bet size. The shaded regions represent the hands for which the caller should call.

Theorem 1.1 (LCP Nash Equilibrium). *The following strategy profile constitutes a Nash equilibrium of LCP:*

$$\begin{aligned}
x_0 &= \frac{3t^2(t-1)}{r^3+t^3-7} \\
x_1 &= \frac{-2r^3+3r^2+t^3-1}{r^3+t^3-7} \\
x_2 &= \frac{r^3+t^3-1}{r^3+t^3-7} \\
x_3 &= \frac{r^3-3r+t^3-4}{r^3+t^3-7} \\
x_4 &= \frac{r^3+3r^2-6r+t^3-4}{r^3+t^3-7} \\
x_5 &= \frac{r^3+t^3+3t^2-7}{r^3+t^3-7} \\
b_0 &= \frac{t^3}{r^3+t^3-7} \\
b(s) &= \frac{t^3(s+1)^3-(3s+1)}{(r^3+t^3-7)(s+1)^3} \\
c(s) &= \frac{r^3+t^3-1+s(r^3+t^3-7)}{(s+1)(r^3+t^3-7)} \\
v(s) &= \frac{r^3+t^3-1+(r^3+t^3-7)(2s^2+4s+1)}{2(r^3+t^3-7)(s^2+2s+1)}
\end{aligned}$$

where $r = L/(1+L)$ and $t = 1/(1+U)$.

Where these threshold values and functions describe the complete strategy profile as outlined in Section ??.

A proof that this profile constitutes a Nash equilibrium is provided in Appendix ??.

It is worth noting the change of variables to (r, t) which significantly simplifies the expressions compared to the original (L, U) formulation. This transformation reveals underlying symmetries and makes many properties more transparent, as we will see in section ??.

It is worth noticing that the strategy profile visually approaches NLCP as $L \rightarrow 0$ and $U \rightarrow \infty$, and approaches FBCP as $L, U \rightarrow B$ for any fixed bet size B . We formalize these convergence results in Section ??.